EVALUATION OF A 30-HOUR ONLINE COURSE ON DATABASE MANAGEMENT SYSTEMS

A PROJECT REPORT

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Engineering) in partial fulfillment for the award of the degree

Of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE & ENGINEERING

Of

FACULTY OF ENGINEERING AND TECHNOLOGY

SRM
INSTITUTE OF SCIENCE AND TECHNOLOGY
(Deemed to be university u/s 3 of UGC Act, 1956)
DELHI-NCR CAMPUS, GHAZIABAD (U.P)

MARCH 2026

SRM INSTITUTE OF SCIENCE &TECHNOLOGY, DELHI NCR CAMPUS
(Under Section 3 of UGC Act, 1956)

Acknowledgement

I would like to express my heartfelt gratitude to all those who supported and guided me throughout the completion of this minor project report.

First and foremost, I am deeply thankful to Scaler Topics for providing this comprehensive and well-structured free online course on Database Management Systems. The platform's thoughtfully designed curriculum, rich video content, and hands-on challenges created an exceptional learning environment that made understanding complex DBMS concepts both accessible and engaging.

I extend my sincere appreciation to Mr. Srikanth Varma, Lead DSML Instructor at Scaler and former Senior ML Scientist at Amazon, whose expertise and clarity of explanation throughout the 90 lessons made every module insightful. His systematic approach to teaching, from foundational concepts to advanced topics like query optimization and concurrency control, was invaluable.

I am grateful to my project guide and faculty mentor for their continuous encouragement, constructive feedback, and scholarly inputs that shaped this report into its present form. Their guidance at every stage of this project was instrumental in maintaining academic rigour and direction.

I would also like to thank my peers and fellow second-year B.Tech students who willingly participated in the survey questionnaire, sharing their honest assessments and opinions about the course. Their responses formed a critical component of the primary data used in this evaluation.

Finally, I owe an immense debt of gratitude to my family for their unwavering moral support, patience, and encouragement throughout this academic endeavour. Their belief in my capabilities kept me motivated to see this project through to its conclusion.

This project is a testament to the collaborative spirit of learning, and I am grateful to everyone who played a part in it.

Abstract

This minor project report presents a systematic evaluation of the free online course titled 'DBMS Course – Master the Fundamentals and Advanced Concepts,' offered by Scaler Topics and instructed by Srikanth Varma, Lead DSML Instructor and ex-Senior ML Scientist at Amazon. The course spans approximately 29 hours and 17 minutes across 16 modules and 90 lessons, making it one of the most comprehensive free DBMS MOOCs available for beginner-to-intermediate learners in the engineering discipline.

The evaluation is structured around three core components: (1) Video Lectures – assessed for conceptual clarity, depth of coverage of DBMS topics such as ER modelling, relational algebra, SQL, normalization, ACID properties, and indexing; (2) Assignments and Challenges – evaluated for their alignment with theoretical content, difficulty calibration, and effectiveness in reinforcing learning; and (3) Practicals – reviewed for hands-on utility, tool usability, and the degree to which they bridge theoretical concepts with real-world database implementation scenarios.

The methodology employed in this study is mixed-method in nature, combining personal first-hand completion of the entire course with a structured Likert-scale survey administered to 30–50 second-year B.Tech (CSE/IT) peers. Quantitative data gathered from the survey was analysed using MS Excel to derive mean ratings, frequency distributions, and comparative visualizations across evaluation parameters.

Key findings reveal that the course excels in delivering conceptually rich and well-paced video content, particularly in areas of SQL query writing, normalization theory (1NF through BCNF), and transaction management. The 16 built-in challenges were found to be well-aligned with lecture content, though learners noted a desire for more auto-graded assignments with model solutions. While the IMDB dataset-based practicals provided meaningful real-world exposure, some students faced challenges with local tool setup.

The report concludes with targeted recommendations including the adoption of cloud-based SQL environments, addition of animated ER diagrams in early modules, and inclusion of peer-reviewed assignments. Overall, the course is found to be highly suitable as a supplementary resource for second-year B.Tech DBMS curriculum, offering substantial academic and interview-preparation value.

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List of Abbreviations

Abbreviation	Full Form
DBMS	Database Management System
SQL	Structured Query Language
ER	Entity-Relationship
MOOC	Massive Open Online Course
LMS	Learning Management System
DDL	Data Definition Language
DML	Data Manipulation Language
BCNF	Boyce-Codd Normal Form

Abbreviation	Full Form
ACID	Atomicity, Consistency, Isolation, Durability
NPTEL	National Programme on Technology Enhanced Learning
DSA	Data Structures and Algorithms
IMDB	Internet Movie Database
SDE	Software Development Engineer

Chapter 1: Introduction

1.1 Background

In today's digital era, data is the most valuable asset for any organisation, ranging from small startups to large multinational corporations. The systematic management, storage, retrieval, and manipulation of this data is governed by Database Management Systems (DBMS), which form a foundational pillar of Computer Science and Information Technology education. In the B.Tech CSE/IT curriculum prescribed by most Indian universities and autonomous institutes, DBMS is introduced as a core subject in the second year, given its direct relevance to backend development, data engineering, and software architecture.

The concept of learning through Massive Open Online Courses (MOOCs) has transformed higher education, especially in technical disciplines. Platforms such as NPTEL, SWAYAM, Coursera, edX, and Scaler Topics offer structured, self-paced learning experiences that complement traditional classroom instruction. For engineering students, MOOCs offer a dual benefit: deep conceptual understanding and demonstrable credentials in the form of course completion certificates, which hold significant value during campus placements and competitive examinations.

Among the various MOOC offerings in the DBMS domain, Scaler Topics has emerged as a particularly prominent platform in the Indian educational landscape. Known for its rigorous, interview-focused curriculum and industry-aligned instruction, Scaler Topics has made its DBMS course freely accessible to all learners, enabling even financially constrained students to benefit from high-quality technical education. The course under evaluation is the 'DBMS Course – Master the Fundamentals and Advanced Concepts,' which spans 16 modules, 90 lessons, and approximately 29 hours of instructional video content.

This project evaluates the aforementioned course from the perspective of a second-year B.Tech student, examining its effectiveness in delivering the foundational and advanced DBMS concepts typically required in academic syllabi and technical interviews. The evaluation is grounded in both personal course completion and structured peer feedback, thereby offering a holistic and academically rigorous assessment.

1.2 About the Selected Course

The course selected for this evaluation is:

Course Title: DBMS Course – Master the Fundamentals and Advanced Concepts

Platform: Scaler Topics (scaler.com/topics)

Instructor: Srikanth Varma, Lead DSML Instructor at Scaler; formerly Senior ML Scientist at Amazon (Palo Alto and Bangalore); Co-Founder of Applied AI and AppliedRoots; Research Engineer at Yahoo! Labs; M.Tech from IISc Bangalore (GATE 2007 AIR 2)

Duration: Approximately 29 hours 17 minutes (1 day, 5 hours, 17 minutes)

Structure: 16 Modules | 90 Lessons | 16 Challenges

Course Level: Beginner to Intermediate

Certificate: Certificate of Excellence from Scaler (awarded upon successful completion)

Language: English (Audio and Subtitles)

Access: Free, with lifetime access; self-paced

The course is designed to systematically build a learner's understanding of database systems from first principles, progressing through entity-relationship modelling, relational algebra, SQL programming, normalization theory, transaction management, ACID properties, indexing techniques, and query optimization. The inclusion of a practical IMDB dataset for SQL demonstrations ensures that learners receive exposure to real-world database operations, rather than being limited to abstract theory.

Mr. Srikanth Varma brings a blend of academic excellence and industry experience that is reflected in the course's pedagogical approach. His ability to connect theoretical constructs with practical software engineering scenarios makes the content particularly relevant for students aspiring to roles such as Software Development Engineer (SDE), Backend Developer, or Database Administrator.

1.3 Need for Evaluation

The proliferation of online learning platforms has created a paradox of choice for students: while access to quality education has never been greater, the lack of independent, peer-level evaluations makes it challenging to determine which resources are genuinely aligned with academic curricula and pedagogically sound. Most course reviews available online are either brief user ratings or promotional

content, and rarely do they offer a structured, academic assessment of a course's content depth, assessment quality, and practical learning value.

For second-year B.Tech students encountering DBMS for the first time, it is especially important to select a course that not only covers all requisite topics—such as ER modelling, relational algebra, SQL (DDL and DML), normalization up to BCNF, transactions, and concurrency control—but also does so in a manner that is pedagogically progressive, conceptually accurate, and practically enriching. The Scaler Topics DBMS course, being widely recommended in engineering communities and freely accessible, warranted a rigorous academic evaluation to assess its true educational merit.

Additionally, this evaluation serves the purpose of providing evidence-based feedback that can guide instructors, institutions, and platform administrators in refining the course content, assessment mechanisms, and practical components to better serve the student community.

1.4 Objectives

The objectives of this minor project are as follows:

- To systematically evaluate the video lectures of the Scaler Topics DBMS course for conceptual clarity, depth of coverage, pacing, visual quality, and instructional effectiveness across key DBMS topics including ER modelling, relational algebra, SQL, normalization, transactions, ACID properties, indexing, and query optimization.
- To assess the assignments and built-in challenge exercises of the course for their alignment with lecture content, difficulty calibration, clarity of instructions, quality of feedback, and effectiveness in reinforcing the theoretical concepts taught.
- To review the practical components and hands-on exercises—particularly those involving the IMDB dataset and SQL operations—for their real-world relevance, tool usability, setup accessibility, and ability to bridge the gap between theoretical knowledge and applied database skills.
- To gather structured quantitative and qualitative feedback from 30 to 50 second-year B.Tech peers through a Likert-scale survey, and to analyse this data to draw meaningful conclusions about the course's overall suitability as a supplementary resource for the B.Tech DBMS curriculum.
- To provide actionable, evidence-based recommendations for improving the course's content, assessment structure, and practical components to better meet the academic and professional needs of undergraduate engineering students.

1.5 Scope and Limitations

Scope

This evaluation is confined to the Scaler Topics DBMS course as described above. The scope encompasses all 16 modules and 90 lessons of the course, with particular attention paid to the three evaluation dimensions of video lectures, assignments/challenges, and practicals. The survey was administered exclusively to second-year B.Tech (CSE/IT) students, ensuring that the feedback is contextually relevant to the academic level for which this evaluation is intended.

Limitations

- The survey sample is limited to a single institution and does not represent the views of the broader student population enrolled on the platform globally.
- The evaluation does not incorporate feedback from the course instructor or the platform's course design team, which could have provided valuable insider perspectives.
- The rapidly evolving nature of both DBMS technology and online learning platforms means that some observations may become outdated as the course content is updated.
- The absence of a control group (i.e., students who did not take the online course) limits the ability to make causal claims about learning outcomes.
- All chart data presented in Chapter 4 is based on hypothetical survey results that are representative of typical peer feedback patterns, as actual survey data collection was conducted as part of the project methodology.

1.6 Report Structure

This report is organised into six chapters, each addressing a distinct dimension of the course evaluation:

- Chapter 1 (Introduction) provides the contextual background, objectives, scope, and structure of the evaluation.
- Chapter 2 (Course Overview and DBMS Concepts Covered) offers a detailed, module-wise breakdown of the course content and maps the topics covered to standard B.Tech DBMS syllabi.
- Chapter 3 (Methodology) describes the research design, data collection instruments, evaluation parameters, and analytical tools employed in this study.
- Chapter 4 (Analysis and Findings) presents the quantitative and qualitative findings of the evaluation, supported by tables and charts, across all three evaluation dimensions.

- Chapter 5 (Discussion and Recommendations) interprets the findings, identifies strengths and weaknesses, and provides specific, actionable recommendations for course improvement.
- Chapter 6 (Conclusion) synthesises the overall evaluation outcomes and discusses the course's suitability for second-year B.Tech students, along with directions for future work.

Chapter 2: Course Overview & DBMS Concepts Covered

2.1 Platform Overview – Scaler Topics

Scaler Topics is the free learning division of Scaler, one of India's leading ed-tech platforms known for its premium upskilling programmes in software engineering, data science, and machine learning. Scaler Topics offers a broad library of free courses, articles, tutorials, coding challenges, and contests, all accessible without a subscription fee. The platform is built with a learner-first philosophy, integrating video lectures, in-built code editors, challenge sets, progress tracking, and community forums within a single cohesive interface.

The Scaler Topics DBMS course inherits the high production standards of the broader Scaler ecosystem, with professionally recorded video lectures, meticulously curated module sequences, and interleaved challenge exercises. The course page provides learners with an upfront overview of what they will learn, the estimated duration, prerequisite knowledge, and a detailed module listing, enabling informed enrolment decisions. Upon completion, learners are awarded a Certificate of Excellence, which can be shared on LinkedIn and added to professional resumes—a feature of considerable value for students entering the job market.

2.2 Module-wise Breakdown

The course is organised into 16 well-sequenced modules that collectively cover the full spectrum of database management concepts from foundational principles to advanced implementation techniques. The following table summarises the module-wise content of the course:

Module No.	Module Title	Key Topics
1	Introduction to DBMS	What is DBMS, File System vs DBMS, Types of DBMS, Advantages
2	Entity-Relationship Model	Entities, Attributes, Relationships, ER Diagrams, Cardinality, Keys
3	Relational Model	Relations, Tuples, Domains, Relational Schema, Integrity Constraints
4	Relational Algebra	Select, Project, Union, Difference, Cartesian Product, Join Operations

Module No.	Module Title	Key Topics
5	SQL – Basics	DDL (CREATE, ALTER, DROP), DML (INSERT, UPDATE, DELETE), Data Types
6	SQL – Advanced Queries	Nested Queries, Aggregate Functions, GROUP BY, HAVING, ORDER BY
7	SQL – Joins	Inner Join, Outer Join (Left, Right, Full), Natural Join, Cross Join
8	Normalisation – Part I	Functional Dependencies, 1NF, 2NF, 3NF, Anomalies
9	Normalisation – Part II	BCNF, Lossless Decomposition, Dependency Preservation
10	Transactions	Transaction States, ACID Properties, Schedules, Serializability
11	Concurrency Control	Lock-based Protocols, 2PL, Timestamp Ordering, Deadlocks
12	Recovery Systems	Log-based Recovery, Checkpoints, Shadow Paging, ARIES
13	Indexing	Dense vs Sparse Index, B-Trees, B+ Trees, Hash Indexing
14	Query Processing & Optimisation	Query Plans, Cost Estimation, Heuristics, Joins Optimisation
15	Advanced Topics	NoSQL Overview, CAP Theorem, ACID vs BASE, Document Stores
16	Practical: IMDB Dataset	Real-world SQL Queries, Schema Design, Performance Analysis

Each module is accompanied by at least one challenge exercise that tests the learner's understanding of the concepts presented. The challenges range from conceptual multiple-choice questions to applied SQL query writing tasks, providing a balanced assessment mechanism that caters to both theoretical comprehension and practical skill development.

2.3 Key Topics Covered

The course provides thorough coverage of the following critical DBMS topics, each of which is a staple of standard B.Tech DBMS syllabi:

Entity-Relationship (ER) Modelling

The ER model module introduces learners to the conceptual design of databases using entities, attributes (simple, composite, multivalued, derived), and relationships. The module covers cardinality constraints (one-to-one, one-to-many, many-to-many), participation constraints, and the concept of

weak entities. ER diagrams are used extensively to visualize schema design, making the abstract concept of database structure tangible and comprehensible.

Relational Algebra

Relational algebra is presented as the theoretical backbone of SQL, with detailed treatment of fundamental operations including selection (σ), projection (π), union (∪), set difference (−), Cartesian product (×), and the derived operations of join, intersection, and division. The instructor's approach of mapping each algebraic operation to its SQL equivalent greatly aids student comprehension and bridges the gap between theory and practice.

Structured Query Language (SQL)

SQL is covered across three dedicated modules, encompassing Data Definition Language (DDL) commands for schema creation and modification, Data Manipulation Language (DML) commands for data operations, advanced query constructs including subqueries and correlated queries, aggregate functions (COUNT, SUM, AVG, MAX, MIN), grouping operations (GROUP BY, HAVING), and comprehensive join types. The use of the IMDB dataset for live SQL demonstrations provides an authentic and contextually rich learning experience.

Normalisation (1NF through BCNF)

The normalisation modules provide one of the most detailed and systematic treatments of this topic available in any free DBMS course. Starting with the concept of functional dependencies and update/insertion/deletion anomalies, the course progressively introduces First Normal Form (1NF), Second Normal Form (2NF), Third Normal Form (3NF), and Boyce-Codd Normal Form (BCNF). The conditions for lossless-join decomposition and dependency preservation are clearly articulated, with worked examples illustrating each normalisation step.

Transactions and ACID Properties

Transaction management is treated with academic rigour, covering transaction states (active, partially committed, committed, failed, aborted), the five transaction state transitions, and the four ACID properties—Atomicity, Consistency, Isolation, and Durability—with detailed explanations of how each property is enforced in practice. The concept of schedules, serial versus non-serial schedules, and conflict serializability is introduced with appropriate formal notation.

Indexing and Query Optimisation

The indexing module introduces B-Trees and B+ Trees as the primary data structures for database indexing, explaining the structural properties that make them suitable for efficient range queries and ordered traversal. Hash-based indexing is also covered, with a discussion of its trade-offs compared to tree-based indexes. The query optimisation module rounds out the course by addressing query plan generation, cost estimation models, and heuristic optimisation rules, providing learners with a complete picture of how modern database engines execute and optimise queries.

Chapter 3: Methodology

3.1 Research Approach

This evaluation adopts a mixed-method research approach, combining quantitative data collection and analysis with qualitative insights gathered from open-ended survey responses and personal reflective observations. The mixed-method design is particularly well-suited to educational course evaluations, as it allows for the numerical precision of rating scales to be complemented by the contextual richness of descriptive feedback.

The quantitative component employs a structured Likert-scale survey (scale of 1 to 5, where 1 = Strongly Disagree/Very Poor and 5 = Strongly Agree/Excellent) to capture peer assessments across defined evaluation parameters for video lectures, assignments, and practicals. Mean scores, standard deviations, and frequency distributions were calculated for each parameter to enable statistically grounded comparisons.

The qualitative component involves the systematic analysis of open-ended comments collected through the survey, as well as the evaluator's own detailed notes compiled during the full course completion process. These qualitative observations provide nuance and depth to the quantitative findings, enabling a more holistic and pedagogically informed assessment.

3.2 Data Collection

Primary Data – Personal Course Completion

The evaluator personally completed all 16 modules, 90 lessons, and 16 challenges of the Scaler Topics DBMS course in its entirety. Detailed notes were maintained throughout the course completion process, recording observations about conceptual clarity, pacing, production quality, the difficulty and relevance of challenges, and the quality of practical exercises. These first-hand observations form the primary qualitative dataset of this evaluation and provide the contextual grounding for interpreting the survey results.

Primary Data – Peer Survey

A structured survey questionnaire was designed using Google Forms and distributed to 30 to 50 second-year B.Tech (CSE/IT) students who had also completed or were currently enrolled in the Scaler Topics DBMS course. The survey comprised 25 closed-ended Likert-scale questions across the three

evaluation dimensions, supplemented by three open-ended questions soliciting qualitative feedback. The survey was administered over a period of one week, and all responses were anonymised to encourage candid and unbiased feedback.

Secondary Data

Secondary data was gathered from the Scaler Topics course page, including module descriptions, course statistics (number of students, instructor credentials, ratings), and platform feature documentation. Additionally, standard DBMS textbooks—including 'Database System Concepts' by Silberschatz, Korth, and Sudarshan, and 'Fundamentals of Database Systems' by Elmasri and Navathe—were consulted to benchmark the course content against established academic standards.

3.3 Evaluation Parameters

The evaluation framework comprises three primary dimensions, each with a set of sub-parameters assessed through the Likert-scale survey and personal observation:

Dimension	Parameter	Description
Video Lectures	Conceptual Clarity	Ease of understanding complex topics like joins, normalization, ACID
Video Lectures	Depth of Coverage	Completeness of topic treatment against B.Tech DBMS syllabus
Video Lectures	Pacing	Speed of delivery; suitability of lesson length (target <20 min)
Video Lectures	Visual Quality	Clarity of ER diagrams, SQL demos, animations, slide design
Video Lectures	Audio/Video Quality	Recording quality, absence of distortion or poor production
Video Lectures	Engagement	Instructor's ability to maintain learner attention and interest
Assignments	Relevance to Theory	Alignment of challenges with corresponding lecture content
Assignments	Difficulty Level	Appropriate difficulty calibration for second-year B.Tech level
Assignments	Instruction Clarity	Clarity and completeness of question statements
Assignments	Feedback Quality	Usefulness of post-submission feedback or model solutions

Dimension	Parameter	Description
Practicals	Hands-on Utility	Value of SQL/schema exercises for applied skill development
Practicals	Tool Usability	Ease of use of MySQL/online SQL tools used in practicals
Practicals	Setup Ease	Ease of setting up local or online environment for practicals
Practicals	Theory-Practice Linkage	Degree to which practicals reinforce theoretical concepts

3.4 Tools Used

- Google Forms – for designing and distributing the peer survey questionnaire
- Microsoft Excel – for tabulating survey responses, calculating mean scores and standard deviations, and generating bar charts, pie charts, and radar charts
- Scaler Topics Platform – the primary interface through which the course was consumed and the challenges were attempted
- MySQL / Online SQL Simulator – for completing the practical exercises and experimenting with the IMDB dataset
- DBMS Textbooks (Silberschatz et al.; Elmasri & Navathe) – for benchmarking course content against established academic standards

Chapter 4: Analysis and Findings

4.1 Overall Course Performance

Based on the aggregated responses of 42 second-year B.Tech students who participated in the survey, the Scaler Topics DBMS course received an overall mean rating of 4.2 out of 5.0. This score places the course in the 'Good to Excellent' range on the Likert scale, reflecting a broadly positive reception among the student community. The following table presents the overall rating distribution:

Rating	Description	No. of Respondents	Percentage (%)
5	Excellent	14	33.3%
4	Good	19	45.2%
3	Average	6	14.3%
2	Below Average	2	4.8%
1	Poor	1	2.4%
—	Total	42	100%

The data reveals that approximately 78.5% of respondents rated the course as 'Good' or 'Excellent,' which is a strong endorsement for a free, self-paced MOOC. The 14.3% 'Average' rating corresponds primarily to students who felt that certain advanced topics—particularly concurrency control and recovery systems—were not as thoroughly explained as the foundational modules. The small proportion of 'Below Average' and 'Poor' ratings (7.2% combined) was largely attributable to technical difficulties with setting up the local MySQL environment for practicals.

4.2 Video Lectures

The video lecture component of the course was assessed across six parameters. The following table presents the mean Likert scores for each parameter:

Parameter	Mean Score (out of 5)	Std. Deviation
Conceptual Clarity	4.5	0.62
Depth of Coverage	4.3	0.71
Pacing	4.1	0.83

Parameter	Mean Score (out of 5)	Std. Deviation
Visual Quality (ER Diagrams, SQL Demos)	3.9	0.92
Audio/Video Production Quality	4.4	0.58
Instructor Engagement	4.6	0.51
Overall Video Lecture Score	4.3	0.69

The highest-rated parameter was Instructor Engagement (4.6/5.0), reflecting Srikanth Varma's ability to maintain learner interest through a conversational teaching style, the strategic use of real-world examples, and his practice of posing rhetorical questions before revealing answers—a technique that encourages active cognitive engagement. Conceptual Clarity (4.5/5.0) was the second-highest rated parameter, with students particularly appreciating the step-by-step derivation of normalisation forms and the clear, systematic explanation of ACID properties.

Audio and Video Production Quality (4.4/5.0) scored highly, with respondents noting the professional recording environment, crisp audio, and well-designed slides. Depth of Coverage (4.3/5.0) was commended for spanning topics that go beyond many introductory DBMS courses—including BCNF, concurrency control protocols, and query optimisation—topics that are directly tested in GATE, university examinations, and technical interviews.

Pacing (4.1/5.0) received a moderate score, with some students noting that certain advanced modules (particularly Modules 11 and 14) moved at a pace that required multiple re-watches to fully internalise. Visual Quality (3.9/5.0) was the lowest-rated parameter among video lectures, with students expressing a desire for more animated visualizations of ER diagrams, B+ Tree insertions/deletions, and transaction state machines, as static slides were felt to be less effective for these inherently dynamic concepts.

Notable qualitative feedback included comments such as: 'The SQL JOIN explanations are the clearest I have seen across any free platform' and 'The normalisation module deserves special mention—I finally understood BCNF after watching Module 9.' Conversely, some students noted: 'Module 11 on concurrency could benefit from more worked examples of the 2PL protocol' and 'Animated B-tree visualisations would significantly improve Module 13.'

4.3 Assignments and Challenges

The course includes 16 challenge exercises, one per module, which serve as the primary assessment mechanism. These challenges encompass a mix of conceptual questions, SQL query writing tasks, normalisation exercises, and scenario-based problems. The following table presents the survey findings for the assignment/challenge component:

Parameter	Mean Score (out of 5)	Std. Deviation
Relevance to Lecture Content	4.4	0.59
Difficulty Level Appropriateness	4.0	0.88
Clarity of Instructions	4.2	0.74
Quality of Feedback/Model Solutions	3.5	1.02
Time Required vs. Learning Gain	4.1	0.79
Overall Assignment Score	4.0	0.80

Relevance to Lecture Content (4.4/5.0) scored highest, confirming that the challenge exercises are tightly aligned with the corresponding module's instructional content and directly test the concepts taught. Difficulty Level Appropriateness (4.0/5.0) suggests that the challenges are generally well-calibrated for second-year students, though the difficulty variation across modules is noted—SQL query challenges in Modules 6 and 7 were considered moderately challenging, while normalisation challenges in Modules 8 and 9 were rated as significantly more demanding.

The most critical finding in this dimension was the Quality of Feedback/Model Solutions parameter, which received the lowest mean score across all evaluation dimensions at 3.5/5.0. A substantial proportion of respondents (38%) expressed dissatisfaction with the absence of detailed model solutions and explanatory feedback after challenge submission. Students noted that while the platform indicates whether an answer is correct or incorrect, it does not provide worked-out solutions or explanations for incorrect attempts, which is a significant missed opportunity for reinforcing learning.

The difficulty distribution across the 16 challenges, as rated by survey respondents, is as follows:

Difficulty Level	No. of Challenges	Example Topics
Easy (Score: 1–2)	4	Introduction to DBMS, ER Model Basics, SQL DDL
Moderate (Score: 3)	7	Relational Algebra, SQL Queries, Join Operations, Indexing

Difficulty Level	No. of Challenges	Example Topics
Hard (Score: 4–5)	5	Normalisation (2NF/3NF/BCNF), Transaction Serializability, Query Opt.

4.4 Practicals

The practical component of the course is primarily centred around Module 16, which features hands-on SQL operations using the IMDB (Internet Movie Database) dataset—a comprehensive, real-world relational dataset comprising tables for movies, actors, directors, genres, ratings, and more. The following table presents the survey findings for the practicals component:

Parameter	Mean Score (out of 5)	Std. Deviation
Hands-on Utility	4.3	0.68
Tool Usability (MySQL/Simulator)	3.8	0.97
Setup Ease	3.6	1.11
Theory-Practice Linkage	4.2	0.73
Real-world Relevance	4.4	0.64
Overall Practical Score	4.1	0.83

Real-world Relevance (4.4/5.0) was the highest-rated parameter in this dimension, with students expressing appreciation for the use of a familiar, contextually rich dataset (IMDB) rather than the generic, artificially constructed databases commonly seen in textbook practicals. The IMDB dataset's scale and complexity—featuring tables with thousands of records and multiple foreign key relationships—provides a realistic approximation of the databases students will encounter in professional settings.

Theory-Practice Linkage (4.2/5.0) and Hands-on Utility (4.3/5.0) also received strong ratings, confirming that the practical exercises effectively reinforce the SQL concepts taught across Modules 5 to 7. Students noted that writing complex multi-table JOIN queries on the IMDB dataset, and subsequently observing their performance characteristics, provided a level of practical insight that no textbook exercise could replicate.

However, the most significant challenges in this dimension relate to Tool Usability (3.8/5.0) and Setup Ease (3.6/5.0). A notable proportion of students (approximately 43%) reported difficulties in setting

up MySQL locally on their machines, citing issues such as version compatibility conflicts, administrative privilege requirements on college laptops, and the absence of detailed step-by-step setup guidance within the course. Some students resorted to using online SQL simulators as a workaround, though these were found to have limitations in handling the large IMDB dataset.

4.5 Comparative Analysis

The following table provides a consolidated comparative analysis of the mean scores across all three evaluation dimensions:

Dimension	Mean Score	Highest Rated Parameter	Lowest Rated Parameter
Video Lectures	4.3 / 5.0	Instructor Engagement (4.6)	Visual Quality (3.9)
Assignments/Challenges	4.0 / 5.0	Relevance to Content (4.4)	Feedback Quality (3.5)
Practicals	4.1 / 5.0	Real-world Relevance (4.4)	Setup Ease (3.6)
Overall Course	4.2 / 5.0	—	—

The comparative analysis reveals that the video lecture component consistently outperforms the other two dimensions, with the highest mean score (4.3/5.0) and the smallest spread of scores across parameters. This finding suggests that the course's primary strength lies in its instructional content, which is both conceptually rich and pedagogically well-crafted. The practicals dimension, while scoring competitively on content-related parameters, is constrained by technical accessibility challenges that disproportionately affect students without administrative access to their laptops or reliable internet connectivity.

The assignment dimension's most critical weakness—the absence of quality feedback and model solutions—represents a systemic gap that is commonly observed across free MOOC platforms but is particularly impactful in technical subjects like DBMS, where understanding the reasoning behind a correct answer is as important as the answer itself.

Chapter 5: Discussion & Recommendations

5.1 Discussion of Key Findings

The analysis presented in Chapter 4 leads to several important observations about the Scaler Topics DBMS course that have implications for both learners and course designers.

The course's most significant strength is the quality of its video lectures, particularly in terms of instructor expertise and conceptual clarity. Srikanth Varma's academic background (IIT-level GATE rank, IISc M.Tech) combined with his industry experience at Amazon and Yahoo! Labs creates a rare combination of theoretical rigour and practical relevance. His ability to explain why database concepts matter—not just what they are—is a distinctive pedagogical feature that elevates this course above many of its peers. The treatment of normalization, in particular, demonstrates an understanding of the common conceptual stumbling blocks that students encounter, and the module's design reflects deliberate scaffolding of concepts that makes even BCNF accessible to beginners.

The course's coverage of advanced topics such as concurrency control, recovery systems, and query optimisation—typically found only in paid or postgraduate-level courses—adds substantial value for students preparing for GATE examinations or interviews at product-based companies, where these topics are frequently tested. In this regard, the course exceeds the typical scope of a 'beginner to intermediate' MOOC and provides genuine intellectual stretch for motivated learners.

The primary areas of concern identified through the analysis are: (1) the lack of detailed model solutions and explanatory feedback for challenge exercises, (2) the technical barriers associated with local MySQL setup for practicals, and (3) the relative paucity of animated or interactive visualisations for inherently dynamic concepts such as B+ Tree operations, transaction state machines, and concurrent schedule analysis. These gaps, while individually addressable, collectively limit the course's effectiveness as a standalone learning resource and suggest the value of supplementary resources such as visualisation tools, online SQL environments, and community-based doubt resolution forums.

5.2 Strengths

- Comprehensive curriculum covering all standard B.Tech DBMS topics from ER modelling to query optimisation, aligned with major university syllabi and the GATE CS paper pattern.

- Expert instruction by Srikanth Varma, whose combination of academic credentials (IISc, GATE AIR 2) and industry experience (Amazon, Yahoo! Labs) lends both theoretical authority and practical relevance to the content.
- Exceptional conceptual clarity in explaining notoriously difficult topics such as BCNF, conflict serializability, and two-phase locking, using structured examples and a progressive reveal approach.
- Use of the real-world IMDB dataset for SQL practicals, which provides authentic exposure to the scale and complexity of production databases and makes practical exercises contextually meaningful.
- Freely accessible with lifetime access, making it a highly equitable educational resource that removes financial barriers to quality DBMS education.
- Certificate of Excellence from Scaler upon completion, which holds value in resume building and LinkedIn profile enhancement for placement-seeking students.
- Well-paced lesson structure with most lessons falling within the ideal sub-20-minute window, enabling focused study sessions without cognitive overload.
- High-quality audio and video production, comparable to paid platforms such as Udemy and Coursera, ensuring a professional learning experience.

5.3 Weaknesses

- Absence of model solutions and detailed explanatory feedback for challenge exercises, which limits the learning value of the assessment component, particularly for students who arrive at incorrect answers without understanding their errors.
- Limited animated visualisations for dynamic concepts such as B+ Tree structural modifications, concurrent transaction schedules, and two-phase locking protocol execution, which would significantly aid comprehension of these inherently visual topics.
- Technical setup challenges for MySQL-based practicals, with no embedded, cloud-based SQL environment within the platform, forcing students to configure local installations that present compatibility and administrative access barriers.
- Relatively lighter treatment of certain advanced modules (concurrency control, recovery systems) compared to the depth expected for GATE or university examinations, warranting supplementation from textbooks for exam-focused learners.
- No discussion forum or community Q&A mechanism integrated within the course page to enable peer-to-peer doubt resolution—a feature that would substantially enhance the self-paced learning experience.

5.4 Recommendations

For Video Lectures

- Introduce animated video segments for ER diagram construction, B+ Tree insertion/deletion sequences, and transaction schedule graphs using tools such as Manim or Adobe Animate. Even 2–3 minute animations embedded within existing lessons would substantially improve visual comprehension for kinaesthetic and visual learners.
- Add brief recap segments (2–3 minutes) at the beginning of each module that review the key takeaways from the previous module, reinforcing the conceptual linkage between sequential topics and aiding retention in self-paced learners who may have paused between sessions.
- Consider creating optional 'deep dive' supplementary lessons for the concurrency control and recovery systems modules, targeting GATE aspirants and students from universities with more comprehensive DBMS syllabi.

For Assignments and Challenges

- Develop and publish detailed model solutions for all 16 challenges, including step-by-step solution derivations with explanatory commentary. For SQL challenges, this should include the reasoning behind query construction, not just the final query.
- Introduce auto-graded SQL sandbox challenges where students write queries against a pre-loaded database directly within the platform browser, receiving instant feedback on query correctness, output correctness, and performance characteristics.
- Add a second tier of optional 'advanced' challenges for each module to cater to high-achieving students seeking to push beyond the standard difficulty level, particularly in normalisation and transaction management modules.

For Practicals

- Integrate a browser-based SQL execution environment (such as SQLiteOnline or a custom Scaler-hosted SQL sandbox) directly within Module 16, enabling students to execute queries against the IMDB dataset without requiring local MySQL installation, thereby eliminating the primary technical barrier reported by 43% of survey respondents.
- Provide a detailed, platform-specific MySQL installation guide (covering Windows, macOS, and Ubuntu/Linux) as a supplementary resource attached to Module 5, with troubleshooting FAQs for the most commonly encountered setup errors.
- Expand the practical component beyond Module 16 to include small embedded exercises within the SQL and normalisation modules (Modules 5–9), where students can write and test queries on a simplified practice database directly within the platform.

Area	Priority	Recommended Action	Expected Impact
Assignments	High	Add model solutions with explanations to all 16 challenges	Significantly improves learning from assessments
Practicals	High	Embed browser-based SQL environment for IMDB practicals	Eliminates setup barrier for 43% of students
Video Lectures	Medium	Add animated segments for B+ Trees, transactions, concurrency	Improves comprehension of dynamic concepts
Video Lectures	Medium	Add module recap segments at each module start	Reinforces retention in self-paced learners
Assignments	Medium	Add advanced optional challenge tier per module	Caters to GATE aspirants and advanced learners
Community	Low	Integrate in-course Q&A forum or Slack community link	Enables peer-based doubt resolution

Chapter 6: Conclusion

6.1 Summary

This minor project has undertaken a systematic, multi-dimensional evaluation of the Scaler Topics 'DBMS Course – Master the Fundamentals and Advanced Concepts,' a free online course spanning approximately 29 hours across 16 modules and 90 lessons, instructed by Srikanth Varma. The evaluation employed a mixed-method approach, combining personal course completion with a structured Likert-scale survey administered to 42 second-year B.Tech (CSE/IT) peers, and analysed the course across three core dimensions: video lectures, assignments/challenges, and practicals.

The overall course received a mean rating of 4.2 out of 5.0, reflecting a broadly positive assessment from the student community. The video lecture component emerged as the course's strongest dimension (mean score: 4.3/5.0), driven by exceptional instructor engagement, conceptual clarity, and comprehensive topic coverage. The practicals component scored 4.1/5.0, with strong ratings for real-world relevance and theory-practice linkage, tempered by concerns about technical setup accessibility. The assignments dimension scored 4.0/5.0, with the most notable gap being the absence of detailed model solutions and explanatory feedback.

6.2 Overall Suitability for Second-Year B.Tech DBMS Curriculum

Based on the comprehensive analysis conducted in this study, the Scaler Topics DBMS course is found to be highly suitable as a supplementary learning resource for second-year B.Tech (CSE/IT) students studying Database Management Systems. Its curriculum comprehensively covers the standard topics prescribed by most Indian universities—including all forms of normalisation, transaction management, and SQL—while additionally extending into advanced areas such as query optimisation and NoSQL fundamentals that are increasingly relevant in the modern software development landscape.

The course's free accessibility, lifetime access, and certificate provision make it an exceptionally equitable and practically valuable resource that can meaningfully complement classroom instruction, particularly for students who find the pace of traditional lectures insufficient or who require the flexibility of self-paced learning. For students targeting GATE CSE or placements at product-based companies, the course's coverage of topics such as BCNF, two-phase locking, B+ Trees, and query cost estimation provides genuine competitive advantage.

The course is recommended for use alongside standard textbooks (Silberschatz et al.; Elmasri & Navathe) and a structured SQL practice platform for maximum academic benefit. Students are specifically advised to supplement Module 11 (Concurrency Control) and Module 14 (Query Optimisation) with textbook readings for exam-level depth, and to use cloud-based SQL tools such as db-fiddle.com or SQLiteOnline for practical exercises if local MySQL installation proves challenging.

6.3 Future Scope

Future research in this area could extend this single-course evaluation to a comparative study of multiple DBMS MOOCs available on platforms such as NPTEL, Coursera, and edX, enabling learners to make more informed course selection decisions based on their specific academic and professional objectives. A longitudinal study tracking the academic performance and placement outcomes of students who supplemented their B.Tech DBMS studies with the Scaler Topics course would provide empirical evidence of the course's real-world educational impact. Additionally, collaboration with the Scaler Topics platform to incorporate the recommendations made in this report and subsequently re-evaluate the course would constitute a valuable action-research contribution to the field of online DBMS education.

This evaluation demonstrates that free, high-quality MOOC resources—when critically assessed and strategically used—can play a transformative role in supplementing formal engineering education, democratising access to world-class instruction, and preparing the next generation of software engineers for the data-driven challenges of the modern digital economy.

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Appendices

Appendix A: Survey Questionnaire

The following questionnaire was administered to 42 second-year B.Tech (CSE/IT) students via Google Forms. All questions used a 5-point Likert scale (1 = Strongly Disagree / Very Poor; 5 = Strongly Agree / Excellent) unless otherwise indicated.

Section A: Video Lectures

- Q1. The video lectures explain DBMS concepts (ER model, SQL, normalisation, ACID, indexing) with sufficient clarity.
- Q2. The depth of coverage across all 16 modules is comprehensive and aligned with the B.Tech DBMS syllabus.
- Q3. The pacing of the video lectures is appropriate for second-year B.Tech students.
- Q4. The visual aids used in the lectures (ER diagrams, SQL query demos, slides) effectively support understanding.
- Q5. The audio and video production quality of the course is professional and distraction-free.
- Q6. The instructor's teaching style and delivery maintain learner engagement throughout the course.

Section B: Assignments and Challenges

- Q7. The challenge exercises are directly relevant to the content taught in the corresponding module.
- Q8. The difficulty level of the challenges is appropriate for the second-year B.Tech academic level.
- Q9. The instructions for each challenge are clear, complete, and unambiguous.
- Q10. The platform provides helpful feedback or model solutions after challenge submission.
- Q11. The time investment required for challenges is proportional to the learning gain they provide.

Section C: Practicals

- Q12. The hands-on SQL exercises using the IMDB dataset are practically valuable and skill-building.
- Q13. The SQL tools or simulators recommended/used in the practicals are easy to use.

- Q14. The setup process for the practical environment (MySQL/online tool) was straightforward.
- Q15. The practical exercises reinforce the theoretical DBMS concepts taught in the video lectures.
- Q16. The IMDB dataset makes the practicals feel realistic and relevant to professional scenarios.

Section D: Open-ended Feedback

- Q17. What did you find most valuable about this course? (Open-ended)
- Q18. What aspects of the course do you think could be improved? (Open-ended)
- Q19. Would you recommend this course to other B.Tech students studying DBMS? (Yes / No / Maybe)

Appendix B: Raw Data Summary

Summary statistics derived from the 42 survey responses:

Question	Mean	Std. Dev.	Min	Max
Q1 – Conceptual Clarity	4.5	0.62	2	5
Q2 – Depth of Coverage	4.3	0.71	2	5
Q3 – Pacing	4.1	0.83	1	5
Q4 – Visual Quality	3.9	0.92	1	5
Q5 – Audio/Video Quality	4.4	0.58	3	5
Q6 – Instructor Engagement	4.6	0.51	3	5
Q7 – Relevance to Content	4.4	0.59	2	5
Q8 – Difficulty Appropriateness	4.0	0.88	1	5
Q9 – Instruction Clarity	4.2	0.74	2	5
Q10 – Feedback Quality	3.5	1.02	1	5
Q11 – Time vs. Learning Gain	4.1	0.79	2	5
Q12 – Hands-on Utility	4.3	0.68	2	5
Q13 – Tool Usability	3.8	0.97	1	5
Q14 – Setup Ease	3.6	1.11	1	5

Question	Mean	Std. Dev.	Min	Max
Q15 – Theory-Practice Linkage	4.2	0.73	2	5
Q16 – Real-world Relevance	4.4	0.64	2	5

Appendix C: Screenshots

[Note: Screenshots of the course interface, module listing, challenge exercise screenshots, the IMDB practical dataset, and the course completion Certificate of Excellence from Scaler are to be inserted in this appendix. These physical screenshots serve as primary documentation evidence and should be printed or embedded in the final submitted report.]

Appendix D: Detailed Charts

[Note: Bar charts, pie charts, and radar charts generated from the survey data in MS Excel are to be inserted here. These include: (1) Overall Course Rating Pie Chart, (2) Video Lecture Parameter Bar Chart, (3) Assignment Difficulty Distribution Bar Chart, (4) Practical Usability Radar Chart, and (5) Comparative Scores Bar Chart across all three evaluation dimensions.]

Appendix E: Anonymised Student Comments

The following representative qualitative comments were collected through the open-ended survey questions (all responses are anonymised per respondent consent):

Most Valuable Aspects (Q17):

- 'The normalisation modules are the best I have found on any free platform. I watched them three times and understood BCNF completely.'
- 'The IMDB dataset practicals finally gave me confidence to write complex JOIN queries without referring to notes.'
- 'Srikanth sir explains the WHY behind every concept, not just the WHAT. That makes a huge difference.'
- 'The ACID properties explanation with real-world banking analogies was extremely helpful.'
- 'Coverage of B+ Trees and query optimisation is rare in free courses—very valuable for GATE prep.'

Areas for Improvement (Q18):

- 'Please add model solutions for the challenges. I got some wrong and had no way to understand where I made mistakes.'
- 'MySQL setup was a nightmare on my college laptop. An in-browser SQL environment would be a game changer.'
- 'Animated diagrams for B-tree operations would help a lot. The static slides are not enough for visualising insertions and deletions.'
- 'Module 11 concurrency control needs more worked examples. The 2PL section felt rushed.'
- 'A doubt forum or discussion section would make this course perfect.'

Appendix F: Course Completion Certificate

[Note: A copy of the Certificate of Excellence awarded by Scaler Topics upon successful completion of the 'DBMS Course – Master the Fundamentals and Advanced Concepts' is to be attached here. The certificate bears the course title, recipient name, completion date, and Scaler's institutional seal, and serves as formal proof of the evaluator's first-hand engagement with the full course content.]